

Ayush Patel

B. Tech (Computer Science and Engineering)| Indian Institute Of Information Technology, Guwahati
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EDUCATION

•Indian Institute Of Information Technology, Guwahati

2024 – Present

B.Tech., Computer Science and Engineering; CGPA: 9.57/10.0

India

•PM Shri Kendriya Vidyalaya, Shahdol

2022 – 2024

XII (CBSE): 97.0% | X (CBSE): 98.2%

India

EXPERIENCE

•Indian Institute of Technology (IIT) Ropar

May 2026 – Jul 2026

Summer Research Intern – Machine Learning

Ropar, India

- Built a production-ready deepfake audio detection system (**VoiceRadar**) in **PyTorch**, achieving **2.544% EER**, **97.46% accuracy**, and **0.9957 ROC-AUC** on an 18,480-file cross-domain test split
- Trained and benchmarked models across four speech corpora – **ASVspoof5 Track 1**, **FakeOrReal**, **In-The-Wild**, and Kaggle Deepfake Audio Detection datasets – to validate cross-domain generalization
- Engineered a physics-informed regularization module for vocal-tract acoustics, an attentive temporal pooling layer, and self-supervised speech backbones (**WavLM-Large**, **HuBERT**) to improve detection robustness

•Amazon ML Summer School

Jul 2026 – Aug 2026

Machine Learning Trainee

Remote

- Completing an intensive curriculum on machine learning fundamentals and deep learning architectures led by Amazon ML engineers, covering supervised learning, neural networks, and model evaluation
- Collaborating with top-ranked engineering students nationwide to solve applied, real-world machine learning problems in a peer-review-driven cohort

SCHOLASTIC ACHIEVEMENTS

- Maintained a **9.57 CGPA** (Current) on a 10.0 scale across 4 semesters at IIIT Guwahati; ranked **Top 5 in Computer Science** departmentwide
- **Amazon ML Summer School 2026**: Selected from 134,000+ applicants nationwide to rank in the top 3,000
- **Engineering Ring of Honour 2025**: Ranked **All India Rank (AIR) 71**, awarded by naukri.com
- **AISSCE 2024 (CBSE Class XII)**: Ranked in the top 1.5% of Kendriya Vidyalaya Sangathan (KVS) students nationwide

PROJECTS

•Audio Deepfake Detection via Green AI

2026

PyTorch, WavLM, Wav2Vec 2.0, Scikit-learn, XGBoost, Python

GitHub

- Built a resource-efficient audio classification framework by extracting frozen self-supervised learning (SSL) embeddings (**Wav2Vec 2.0**, **WavLM**) through a custom **Python CLI**, eliminating the need for costly end-to-end fine-tuning
- Designed a low-carbon detection pipeline using classical ML backends (**SVM**, **XGBoost**), achieving **0.90% EER** with fewer than **1,000 trainable parameters**

•Zivro – AI-Powered Home Service Marketplace

2026

Java 21, Spring Boot 3.4, MySQL, Spring Security, RESTful APIs, Docker

GitHub

- Architected and built a backend **RESTful API** in **Java 21** and **Spring Boot 3.4** for an AI-assisted home service marketplace connecting customers with verified workers
- Implemented secure **JWT-based authentication** with **Spring Security** across three role hierarchies (USER, WORKER, ADMIN) and managed schema versioning with **Flyway** (V1–V6) on a relational **MySQL 8.x** database
- Developed backend business logic for dynamic urgency-based price quoting, location-based worker matching, multi-step booking workflows, and **Razorpay** payment-verification API endpoints

TECHNICAL SKILLS

- **Programming Languages**: Java, Python, C, SQL
- **Backend & APIs**: Spring Boot, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, JWT Authentication
- **Machine Learning & Data Science**: PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn
- **Databases**: MySQL, Flyway (Schema Migrations)
- **Tools & Technologies**: Git, GitHub, Docker, Postman, Maven, IntelliJ IDEA, VS Code, PyCharm, Jupyter Notebook

KEY COURSES & PROFESSIONAL CERTIFICATIONS

- **Computer Science Coursework**: Artificial Intelligence, Operating Systems, Database Management Systems, Computer Organisation and Architecture, Data Structures and Algorithms, IT Workshop I (Java)
- **Certification**: Professional Data Science Certificate – View Credential

POSITIONS OF RESPONSIBILITY

- **Club Coordinator, BioSkoPien (Movie Club)** (*Aug 2025 – May 2026*): Led a 15-member team in planning and executing content development workshops
- **PR Team Member, IIITG MUN** (*Jan 2025 – Mar 2025*): Drove delegate acquisition through strategic outreach campaigns across multiple colleges